AP PRECALCULUS

Unit 1 • Lesson 1.5 practice

Lesson 1.5 • Polynomial Functions and Complex Zeros

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How many complex zeros, counted with multiplicity, does a polynomial of degree 6 have?

(A) 3
(B) 5
(C) 6
(D) 12

Question 2

NO CALCULATOR

What are the zeros of $p(x) = x^3 + 4x$?

(A) $0,\ 2,\ -2$
(B) $0,\ 2i,\ -2i$
(C) $0,\ 4i$
(D) 0 only

Question 3

NO CALCULATOR

A polynomial with real coefficients has zeros 3 and $2i$. What is its least possible degree?

(A) 2
(B) 3
(C) 4
(D) 6

Question 4

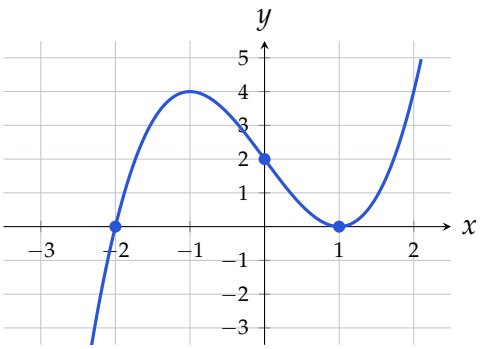
NO CALCULATOR

Which function is odd?

(A) $f(x) = x^4 + x$
(B) $f(x) = x^3 + 1$
(C) $f(x) = x^3 - 5x$
(D) $f(x) = |x|$

Question 5 (graph) NO CALCULATOR

The graph of a cubic polynomial q is shown. It passes through $(0, 2)$. Which could be $q(x)$?



(A) $(x+2)(x-1)^2$
(B) $(x-2)(x+1)^2$
(C) $(x+2)^2(x-1)$
(D) $-(x+2)(x-1)^2$

Question 6 (Free response) NO CALCULATOR

Let $p(x) = (x-1)^2(x+3)(x^2+1)$.

(a) Find the degree of p and all of its complex zeros with their multiplicities.
(b) Describe the behavior of the graph of p at each x-intercept.
(c) Is p even, odd, or neither? Justify your answer.